

# **Transformer-Based Classification of Contract Clause**

## **Themes**

### **Deep Learning Systems Analysis Report**

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## Report Overview

This project created a reproducible deep learning experiment that classifies contract passages into ten clause themes using a custom PyTorch Transformer. The work adapts the expert-annotated Contract Understanding Atticus Dataset (CUAD) into a single-label classification task and compares two otherwise identical models whose only major difference is dropout. The prototype demonstrates a semantic-intelligence capability relevant to ICM Solutions contract review, knowledge reuse, and document-quality workflows while preserving human accountability for legal and procurement decisions.

## Dataset and Task Description

CUAD is a publicly available legal-language dataset created for contract review. The original release contains 510 commercial contracts, more than 13,000 expert-supervised annotations, and 41 clause categories (Hendrycks et al., 2021). The source is appropriate for academic deep learning because it contains real, non-synthetic contract language annotated under legal-expert supervision. This project did not reuse data from an earlier capstone project.

The source annotations were adapted from span-oriented contract review into a ten-class, single-label clause-theme classification task. Cleaning removed exact normalized-text duplicates, conflicting multi-category passages, and records outside the ten selected categories. The final dataset contains 4,356 unique passages from 462 source contracts. Contract-grouped splitting assigned 3,486 passages from 368 contracts to training, 435 passages from 48 contracts to validation, and 435 passages from 46 contracts to testing; no source contract appears in more than one split.

| Clause category | Passages |
|-----------------|----------|
| Audit Rights    | 605      |

|                           |     |
|---------------------------|-----|
| Anti-Assignment           | 556 |
| Insurance                 | 521 |
| Cap On Liability          | 505 |
| Governing Law             | 442 |
| Revenue/Profit Sharing    | 381 |
| Minimum Commitment        | 363 |
| Post-Termination Services | 361 |
| License Grant             | 351 |
| IP Ownership Assignment   | 271 |

The ten classes are not perfectly balanced, but every class is represented in each partition. A vocabulary of 4,417 tokens was created only from training passages. Validation and test tokens absent from that vocabulary map to an unknown-token index, reducing feature-construction leakage. Sequences were padded or truncated to 256 tokens; only eight of the 435 test passages exceeded that limit.

## Model Architecture and Design Decisions

A Transformer was selected because the input is variable-length legal text and the task depends on relationships among words distributed across a clause. The Transformer architecture uses self-attention to model token interactions without recurrence, making it suitable for parallel sequence representation (Vaswani et al., 2017). The project intentionally implemented a compact custom encoder rather than fine-tuning a large pretrained model so that the architecture, parameter initialization, training behavior, and controlled comparison remained transparent.

| Component             | Implemented setting                         |
|-----------------------|---------------------------------------------|
| Vocabulary            | 4,417 training-only tokens; <PAD> and <UNK> |
| Maximum length        | 256 tokens                                  |
| Token embedding       | 128 dimensions, learned                     |
| Position embedding    | 128 dimensions, learned                     |
| Transformer encoder   | 2 layers, 4 attention heads                 |
| Feed-forward sublayer | 256 units with GELU                         |
| Pooling               | Padding-mask-aware mean pooling             |
| Output                | Linear layer with 10 logits                 |
| Trainable parameters  | 864,650                                     |

GELU was used in the feed-forward sublayers because it provides a smooth, input-dependent activation that has performed well across language tasks (Hendrycks & Gimpel, 2016). Padding masks were provided to both self-attention and mean pooling so artificial padding positions did

not influence the passage representation. Cross-entropy loss matched the mutually exclusive ten-class target. AdamW was used with a learning rate of 0.0005 and weight decay of 0.01 because it separates weight decay from adaptive gradient updates (Loshchilov & Hutter, 2019). Gradient norms were clipped at 1.0, the batch size was 64, and both configurations ran all 15 epochs. Macro F1 was the primary checkpoint metric because it gives equal weight to each class instead of allowing the largest classes to dominate the comparison. This is appropriate when performance on less frequent classes remains operationally important (scikit-learn developers, 2026). Lower validation loss broke any macro-F1 tie. Accuracy and weighted F1 were retained as complementary measures.

## Experimental Comparison

The controlled experiment changed exactly one major regularization setting: dropout increased from 0.10 in the baseline to 0.30 in the experimental model. All data records, split assignments, vocabulary, maximum length, model dimensions, initial parameter values, training-batch order, loss function, optimizer, learning rate, weight decay, gradient clipping, batch size, number of epochs, and checkpoint rule remained fixed. An explicit equivalence audit verified matching parameter names, shapes, initial tensors, and first shuffled batch before formal training. Dropout was selected because both pilot and full training could reveal whether stronger regularization reduced the observed gap between near-perfect training performance and lower validation performance. The comparison was not designed to search many hyperparameters or maximize test performance. The test split remained untouched until both validation-selected checkpoints were frozen.

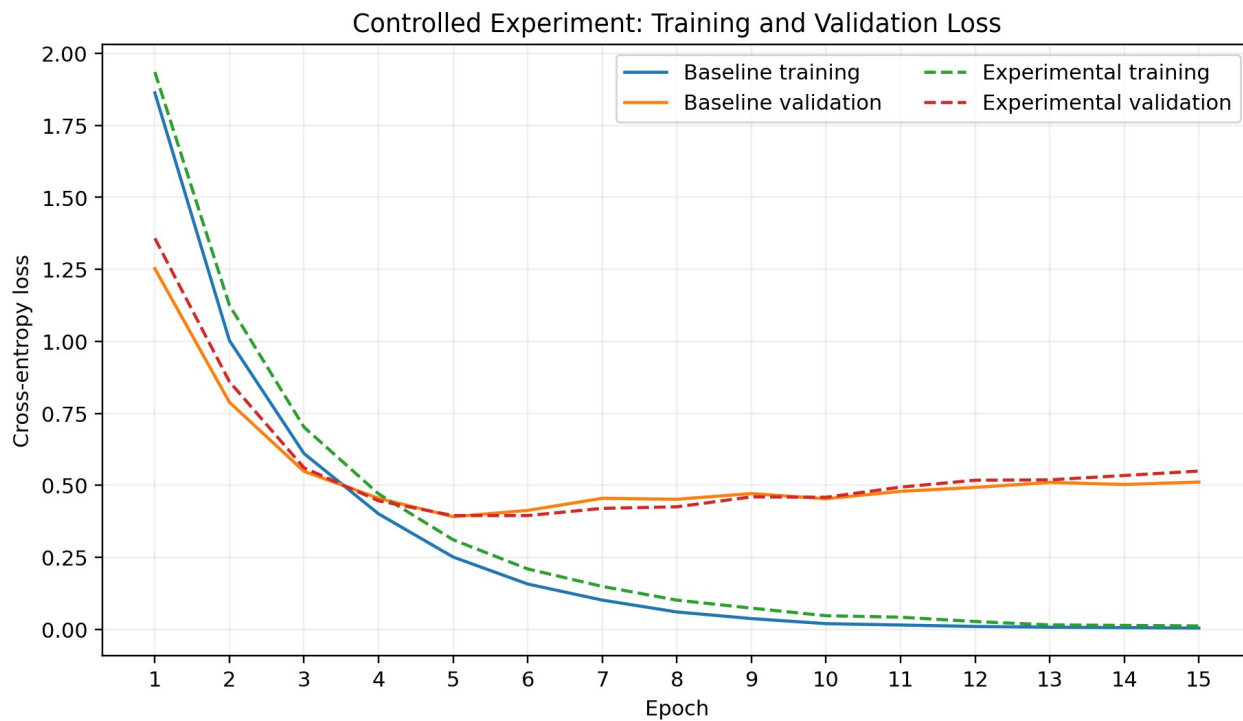


Figure 1. Training and validation loss for the controlled 15-epoch experiment.

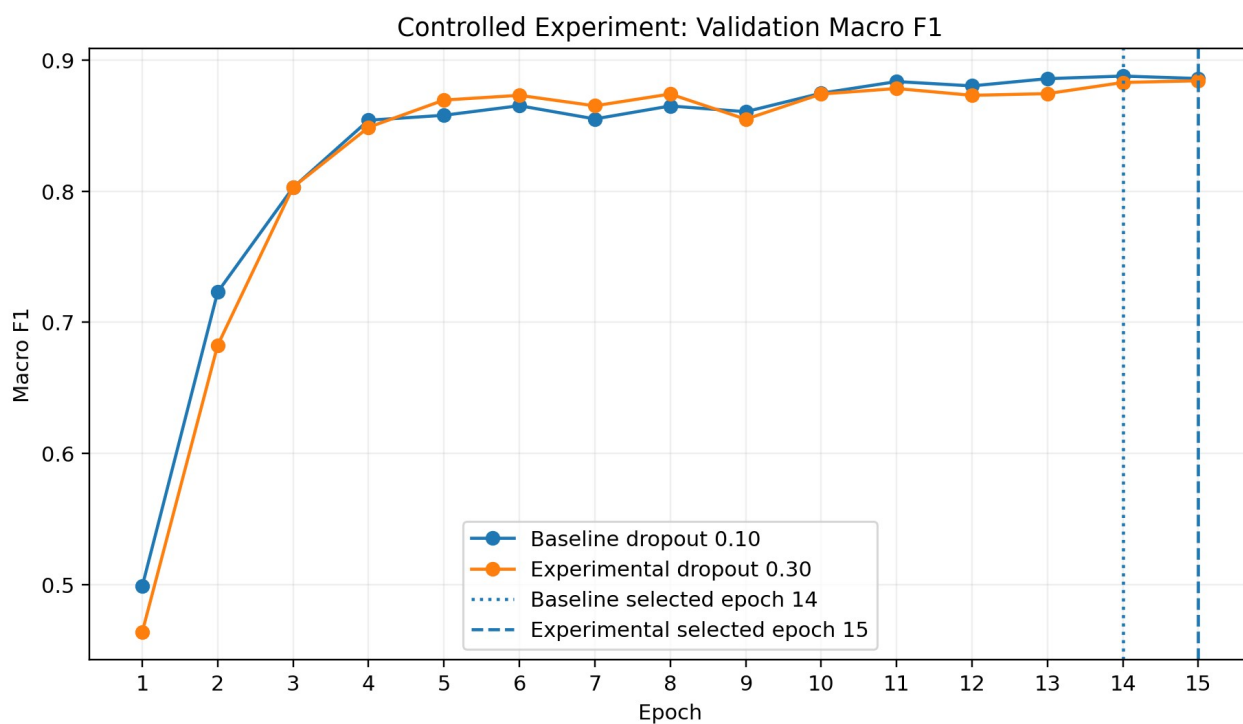


Figure 2. Validation macro F1 and the selected checkpoint epoch for each configuration.

## Results and Interpretation

The baseline reached its highest validation macro F1 of 0.8878 at epoch 14, with validation accuracy of 0.9011 and loss of 0.5029. The experimental model reached its highest validation macro F1 of 0.8842 at epoch 15, with accuracy of 0.8966 and loss of 0.5494. Because the predeclared checkpoint rule favored the higher macro F1, validation selected the baseline.

The untouched test set produced a modest reversal. The experimental configuration achieved test macro F1 of 0.8530, compared with 0.8471 for the baseline, a difference of 0.0059. Experimental accuracy was 0.8736 versus 0.8690, and weighted F1 was 0.8689 versus 0.8628. The baseline nevertheless retained the lower test loss, 0.6410 versus 0.7086. These small, mixed differences support a cautious conclusion: dropout 0.30 produced comparable overall performance and a slight test advantage, but this single-seed experiment does not establish that stronger dropout is reliably superior.

| Metric          | Baseline 0.10 | Experimental 0.30 | Difference |
|-----------------|---------------|-------------------|------------|
| Test loss       | 0.6410        | 0.7086            | +0.0676    |
| Accuracy        | 0.8690        | 0.8736            | +0.0046    |
| Macro precision | 0.8672        | 0.8693            | +0.0021    |
| Macro recall    | 0.8448        | 0.8500            | +0.0052    |
| Macro F1        | 0.8471        | 0.8530            | +0.0059    |
| Weighted F1     | 0.8628        | 0.8689            | +0.0061    |

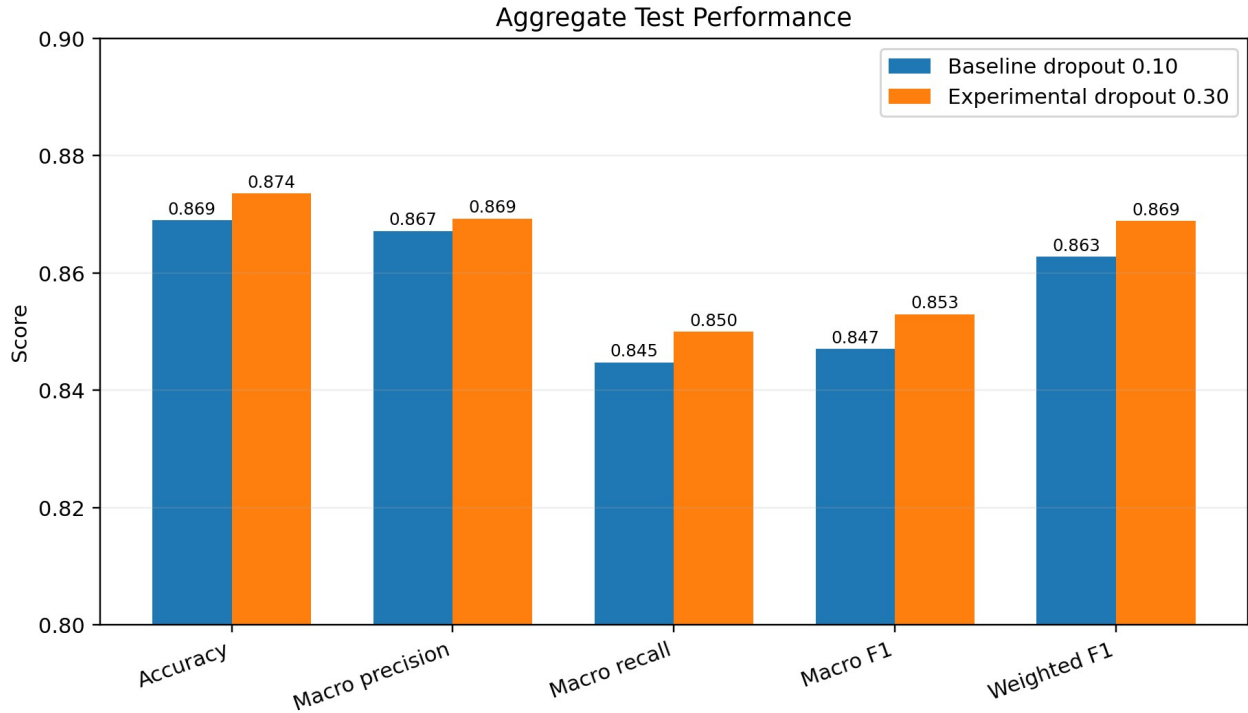


Figure 3. Aggregate test metrics for the frozen validation-selected checkpoints.

The most important training behavior was overfitting. Training loss continued toward zero for both models, while validation loss reached its minimum near epochs 5-6 and then generally increased. Validation macro F1 continued to fluctuate upward slightly despite the rising loss, which explains why the macro-F1 checkpoint rule selected late epochs. Dropout 0.30 slowed fitting but did not materially remove the train-validation gap: at the selected epochs, the macro-F1 gaps were 0.1110 for the baseline and 0.1140 for the experimental model.

Class-level analysis revealed stronger performance on highly distinctive categories such as Insurance and Governing Law. The experimental model achieved F1 of 1.000 for Governing Law, while both models achieved recall of 1.000 for Insurance. The weakest class was IP Ownership Assignment, with F1 of 0.585 for the baseline and 0.622 for the experimental model. Post-Termination Services was also difficult, with F1 of 0.738 and 0.742.

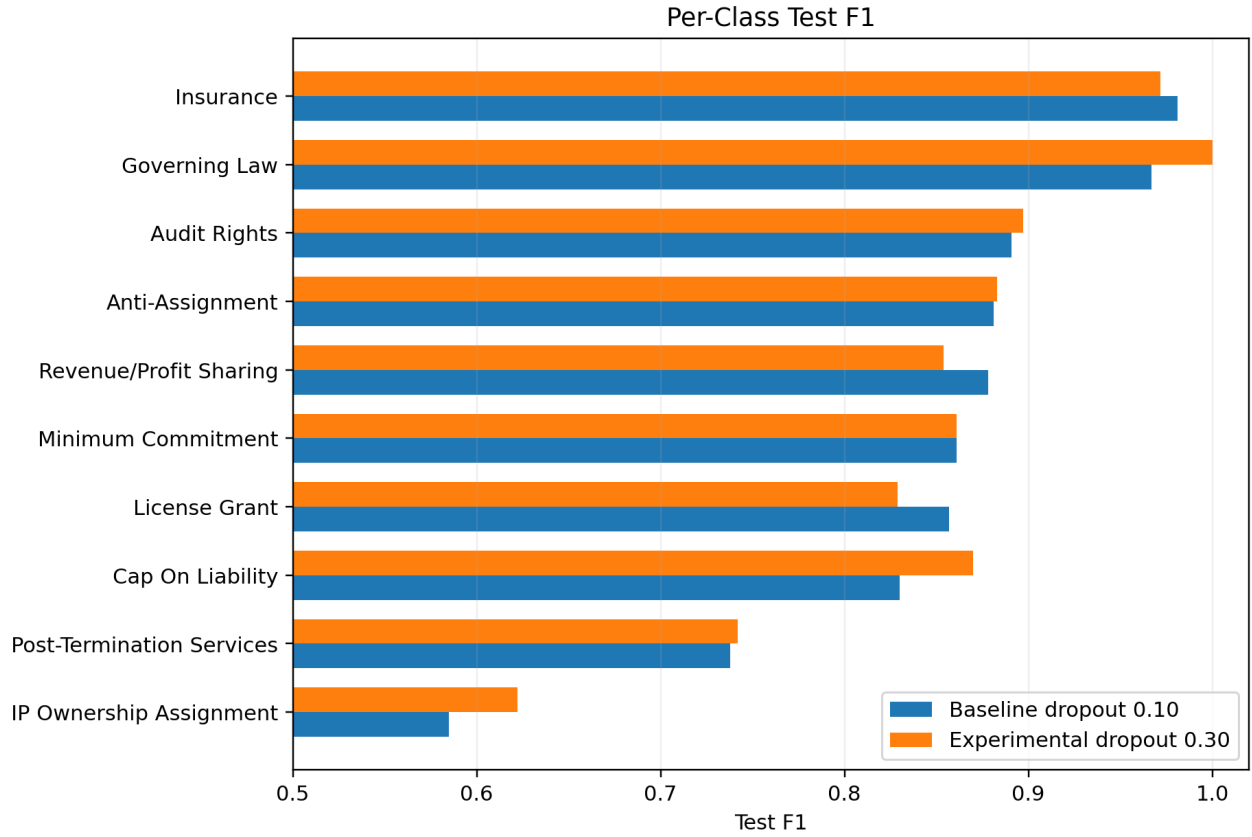


Figure 4. Per-class test F1 for the baseline and experimental configurations.

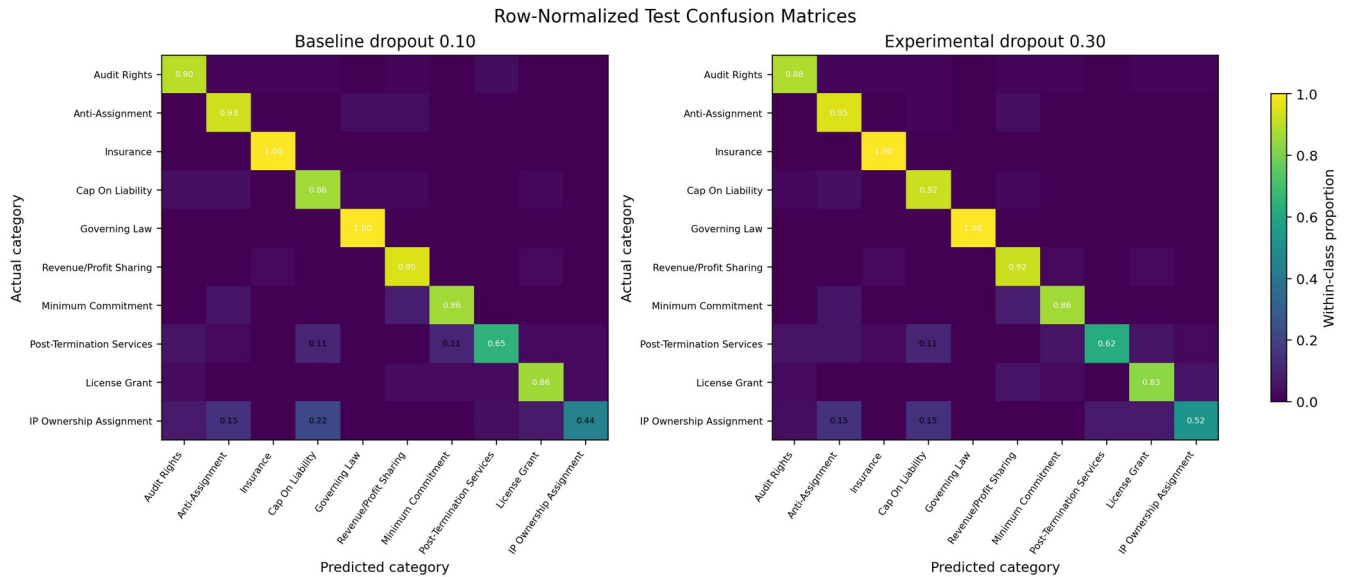


Figure 5. Row-normalized test confusion matrices for both configurations.



A concrete high-confidence error illustrates why accuracy alone is insufficient. Both models assigned approximately 1.000 confidence to an Audit Rights prediction for a passage labeled Post-Termination Services. The passage stated that records must remain available for inspection and audit for one year after expiration or termination. The models focused on the explicit audit vocabulary and missed that the annotation emphasized the obligation after termination. Another recurring pattern involved IP Ownership Assignment clauses being predicted as Cap on Liability or Anti-Assignment when the passages contained overlapping words such as transfer, assignment, liability, and obligations.

The two configurations produced different predictions on 20 of 435 test passages, and exactly one model was correct in 16 of those disagreements. This finding shows that the dropout change affected specific boundary cases even though aggregate scores remained close. Only eight test passages were truncated; baseline accuracy on that subgroup was 0.625 and experimental accuracy was 0.500. The subgroup is too small for a stable conclusion, but it supports investigating longer-context or hierarchical models.

For a non-technical reader, the model correctly identified roughly 87% of held-out clause passages, but some categories with overlapping legal language remained difficult. The small difference between configurations is less important than the demonstrated workflow: contract-grouped data controls, a one-variable experiment, transparent checkpoint selection, and direct review of errors. For ICM Solutions, the output could help prioritize passages for human review or connect clauses to prior templates and quality checks, but it should not decide what a contract means or whether ICM should accept a term.

## Limitations and Risks

The dataset is not an ICM-specific or public-sector contracting corpus. CUAD primarily represents commercial contracts associated with corporate transactions, so vocabulary, clause structure, document quality, and annotation priorities may differ from California solicitations, state contracts, subcontracts, or IV&V deliverables. Only ten of the original 41 categories were modeled, and the selection favored categories with sufficient clean examples. Performance therefore cannot be generalized to all contract-review questions.

The experiment used one random seed and one contract-grouped split. Repeated-seed training or cross-validation could reveal that the 0.0059 test macro-F1 difference falls within ordinary training variation. The test set contained only 435 passages, including 27 IP Ownership Assignment passages and eight truncated passages. Per-class and subgroup estimates for the smaller categories are consequently uncertain.

The tokenizer was trained from scratch and does not use pretrained legal-language representations. A 256-token maximum discards the tail of long passages, and padding-aware mean pooling compresses the sequence into one vector that may lose localized distinctions. The model also produced near-certain probabilities for incorrect predictions, so its softmax scores are not calibrated measures of legal certainty.

Both models overfit the training data. Although the notebook restored validation-selected checkpoints, the formal experiment intentionally ran all 15 epochs without early stopping to preserve the controlled design. In a production-oriented system, continued training after validation loss begins increasing would add cost and may worsen generalization. Model behavior could also drift if contract language, business domains, or document-extraction quality changes.

## Ethical and Responsible Use

The central misuse risk is treating a classification as legal advice or as an authoritative interpretation of contractual rights. The observed high-confidence errors show that a model can be wrong even when its probability appears decisive. Outputs should therefore be framed as routing or review assistance, with the original clause, predicted theme, uncertainty, and relevant precedent available to a qualified human reviewer.

Training-data coverage can create systematic bias. CUAD reflects particular commercial-contract types and expert labeling practices; categories or phrasing common in government procurement, small-business subcontracting, public benefits systems, or ICM client work may be underrepresented. Applying the model without domain validation could cause reviewers to overlook clauses that use unfamiliar language or to over-prioritize classes that CUAD defines more clearly.

Operational use would also require confidentiality controls. Real ICM or client contracts may contain nonpublic pricing, personnel, security, intellectual-property, and negotiation information. A production workflow should limit data access, retain audit logs, document model and dataset versions, test error rates by document type, monitor drift, and define escalation rules. The NIST AI Risk Management Framework emphasizes governing, mapping, measuring, and managing risk throughout the AI lifecycle rather than relying on a one-time accuracy result (National Institute of Standards and Technology, 2023).

This prototype is appropriate only as human-in-the-loop decision support. It should not autonomously accept or reject terms, determine compliance, score counterparties, make procurement decisions, or replace counsel. ICM remains responsible for validating the complete contract, governing law, client requirements, and business context before taking action.

## Future Improvements

The first improvement should be repeated-seed evaluation and uncertainty reporting. Training each configuration across several seeds would show whether the small test difference is stable. Early stopping or a maximum-epoch rule based on validation loss could then be evaluated in a separate controlled experiment, rather than inferred from the present run.

Modeling improvements could include a pretrained legal-language encoder, subword tokenization, calibrated probabilities, class-aware loss functions, and longer-context or hierarchical passage encoders. These changes should be tested one at a time against the same contract-grouped split and should retain per-class and error-level reporting. Calibration is especially important because the current models produced high-confidence errors.

The most valuable domain improvement would be a human-reviewed ICM/public-sector contract corpus. A pilot could begin with nonconfidential sample clauses labeled by ICM subject-matter reviewers using a documented taxonomy. Cross-domain testing should measure whether the CUAD-trained model transfers to public-sector agreements and identify categories that require new labels or examples.

Within the capstone roadmap, the classifier can become one component of a grounded knowledge-reuse system. A later generative-AI project could retrieve relevant clauses, prior findings, templates, and policy guidance based on the predicted theme, while an agentic workflow could route low-confidence or high-risk cases to designated reviewers. The classifier should remain observable and reversible, with retrieved evidence and human approval required before any output affects a contract or pursuit decision.

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